3. DATA

This project combines four datasets, all publicly available through NYC OpenData and accessible using their respective APIs. Together, they document different mobility frictions across New York City.

311 Service Requests: Published by the NYC Office of Technology and Innovation, this dataset lists all service requests from 2020 to the present, updated daily. The unit of observation is an individual service request, with more than 40 columns describing each complaint. Relevant to this project are locational information, complaint type, and additional details about the reported problem. As of the writing of this paper, the dataset contains approximately 22.8 million observations, although significantly fewer are relevant to mobility.

Motor Vehicle Collisions: Published by the NYPD, this dataset documents police-reported motor vehicle collisions involving more than $1,000 in damages. The unit of observation is an individual collision. To remain consistent with the timeframe of the 311 dataset, I limit observations to 2020 onward, leaving approximately TKTK collisions. Of the dataset's 29 columns, particularly relevant are locational information, crash dates, and the number of people injured or killed.

Parking Violations Issued: Published by the Department of Finance, eight datasets are considered one for the purposes of this project, originally divided by fiscal year (included are fiscal years 2020-27). Together, they contain approximately 115 million observations, each representing an issued parking violation. Relevant variables include the location and type of violation. In particular, I am interested in violations that may indicate obstacles to movement, such as double parking or parking in bicycle lanes.

Closed Street Pothole Work Orders: Published by the Department of Transportation, this dataset documents completed pothole repairs across NYC. The unit of observation is a closed pothole work order, with approximately 412,000 observations. Relevant variables include the location of the repair and the date it was completed. Unlike 311 complaints, which record reported problems, these observations document maintenance performed by the city.

Significant work is necessary to turn these datasets into meaningful measures of mobility friction. The 311 dataset, for example, contains all types of complaints, including many unrelated to mobility (noise, building maintenance, or general questions, for example). To identify relevant observations, selected 31 complaint types that plausibly affect movement through the city, including, but not limited to, illegal parking, street condition, abandoned vehicle, street light condition, derelict vehicles, traffic signal condition, and sidewalk condition (these complaint types are listed in order of frequency, from most to least). Similar decisions are necessary for the other datasets, like parking violations, for example, where not every violation necessarily represents an obstacle to movement.

Another consideration is that not all mobility frictions affect travelers equally. A damaged sidewalk, for example, is of concern for pedestrians, while highway conditions matter much more to drivers. Other problems, such as illegally parked vehicles or broken traffic signals, may

affect multiple modes of transportation. I therefore classify the relevant observations by the mobility mode they affect (pedestrians, cyclists, drivers, or some combination of the three).

The data is, in itself, both accessible and relatively clean. What makes my project interesting is not the difficulty of using the data, but the sources I am combining in an effort to think holistically about what it is like to travel a specific street. Still, there are important limitations to what story these records can tell. A 311 complaint documents a reported problem, rather than necessarily the true prevalence of that problem. Similarly, parking violations reflect enforcement activity, not every instance of illegal parking, while closed pothole work orders document repairs rather than every pothole that exists. These distinctions are especially important when comparing neighborhoods, where differences in reporting or enforcement may produce differences in the data that do not reflect actual street conditions. I hope to account for these limitations as I construct the measures, recognizing that the resulting dashboard will be an approximation of mobility friction rather than a perfect representation of every traveler's experience.

This project combines five datasets, all publicly available on NYC OpenData, using their respective API key. The backbone of the mobility frictions are sourced from the 311 service requests published by NYC Office of Technology and Innovation. The dataset contains all service requests from 2020 to present, updated automatically daily. The unit of observation is a service request, and there are more than forty rows. Relevant to this project are locational information, complaint type, and potential details. As of the writing of this paper, 22.8 million rows are recorded, although significantly less are considered relevant. Another included dataset are motor vehicle collisions, published by the NYPD. The unit of analysis all police reported

motor vehicle collisions with damages of more than $1000. To be consistent with the timeframe of the 311 dataset, car crashes start in 2020, which includes TKTK observations. Of the 29 columns, locational data and fatality data are included. Next, eight datasets are considered one, originally dividing parking violations by fiscal year (2020-2027). These datasets are published by the Department of Finance, containing in total 115 million observations, the unit of which being an issued violation. Locational data and violation type are included. Next, a list of closed street pothole work orders published by the Department of transportation. Locational data and date are relevant. The dataset has 412k observations, the unit of which is a closed pothole.

Significant work was done to turn these datasets into lists of mobility frictions. The 311 dataset, for example, contains all types of complaints, including those not related to mobility (noise, building codes, etc). 31 complaint types are considered relevant to this project, including, but not restricted to: Illegal Parking, Street Condition, Abandoned Vehicle, Street Light Condition, Derelict Vehicles, Traffic Signal Condition, Sidewalk Condition, and more (listed in order of most frequent).

The data is, in itself, both accessible and clean. What makes my project interesting is not the difficulty of using the data, but the sources I am combining in an effort to think holistically about what it is like to travel a specific street. I will also discuss limitations, particularly that 311 measures what people report, rather than necessarily the true existence of a problem.